

DHAARANI GANESH THANGAM

Contact | Email | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Worked on full-stack web and mobile development, agentic AI systems, cloud deployment (AWS) and machine learning applications. Proven track record building scalable, production-grade systems including AI-powered applications using LangChain, RAG pipelines, and LLMs.

EDUCATION

Arizona State University

Master of Science in Computer Software Engineering

Tempe, AZ
Aug 2024 - May 2026

Kumaraguru College of Technology

Bachelor of Engineering in Computer Science and Engineering

Coimbatore, India
Jun 2019 - Apr 2023

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, SQL, JavaScript, HTML, C

Frameworks & Libraries: React.js, FastAPI, Flask, Pydantic, Motor, Jetpack Compose, Node.js, SQLAlchemy, Streamlit, TensorFlow, Pandas, ROS

AI/ML: LangChain, RAG Pipelines, ChromaDB, Vector Databases, Groq LLM, Gemini API, AssemblyAI, VAPI AI

Cloud & DevOps: AWS (EC2, RDS), Vercel, Render, Git/Github, CI/CD, pgAdmin

Databases: PostgreSQL, pgvector, MongoDB Atlas

Tools: VS Code, Android Studio, Arduino IDE, Gazebo, Rviz, Tableau, IBM Cognos Analytics, ANTLR4

Methodologies: Agile/Scrum, Agentic AI Development, REST API Development, Async Programming, Object-Oriented Design, JWT Authentication

PROFESSIONAL EXPERIENCE

Kumaraguru Centre for Industrial Research and Innovation

ROS Software Engineering Intern

Coimbatore, India
Jun 2023 - Dec 2023

- Engineered a virtual Autonomous Ground Vehicle (AGV) using ROS 1 (Melodic), enabling full simulation and testing of navigation algorithms in virtual environments.
- Implemented real-time 2D LiDAR Mapping and SLAM-based localization for dynamic indoor AGV navigation, reducing manual calibration via autonomous environment mapping.
- Executed physical teleoperation for a four-wheel AGV and conducted point-to-point simulation using Rviz and Gazebo; documented control workflows to support team onboarding.

National University of Singapore & Hewlett Packard Enterprise

Data Analytics Intern - Global Academic Internship Programme

Singapore
Jun 2022 - Jul 2022

- Processed and analyzed 30,000+ customer demographic data points to drive segmentation using unsupervised ML clustering techniques.
- Built stock prediction models for 5 S&P 500 companies achieving 80% accuracy and developed a full-stack web app integrating two ML models to deliver personalized portfolio recommendations.

PROJECTS

ProposAI - RFP Generator | React, FastAPI, LangChain, RAG, Groq LLM, Gemini, ChromaDB, MongoDB

May 2026

- Built a full-stack RAG pipeline generating RFP responses in ~30s using ChromaDB, Gemini embeddings, and 5 chained Groq LLM calls.
- Reduced deployment image by ~490 MB by replacing local Sentence Transformers with hosted Gemini embeddings, fitting Render's 512 MB limit.
- Secured APIs with OTP authentication, JWT sessions, rate limiting, and token-based per-user data isolation.
- Added Word export with `python-docx` and protected metered LLM APIs from quota abuse.

Dementia Care Management System | Kotlin, Jetpack Compose, Flask, PostgreSQL, AWS, RAG

Jan 2026 - May 2026

- Developed the full Android application using Kotlin and Jetpack Compose with JWT-based role access control for Patients, Caregivers, Family Members, and System Admins in a 6-member Agile Scrum team.
- Built and managed the PostgreSQL database using pgAdmin and implemented a RAG pipeline (PubMedBERT, pgvector, Gemini) for AI-powered cognitive assessments and VAPI AI voice assistant with AssemblyAI speech analysis.

Secure Drone Command & Control System | Java, OOP, Encrypted Messaging, Authentication

Aug 2025 - Dec 2025

- Designed a Java-based drone control system with modular OOP architecture supporting real-time command execution and state reporting between ground control and drone units.
- Integrated encrypted messaging and authentication to secure control links; validated reliability under 10+ simulated scenarios ensuring robust error handling and system stability.